

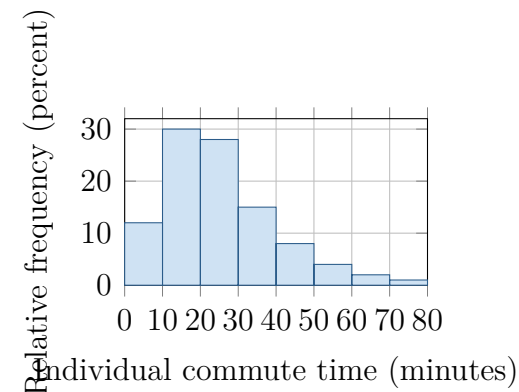
When Is the Mean Nearly Normal?

Unit 2 · Topic 2.12 · TODAY'S PROBLEM

Individual commute times in a very large city are strongly right-skewed, with population mean $\mu = 25$ minutes and standard deviation $\sigma = 15$ minutes. Consider independent random samples.

Rae: “Any average is approximately normal, even for $n = 4$.”

Sol: “With this long right tail, $n = 4$ may be too small; $n = 40$ is more defensible.”



FIRST TAKE (5 min, on your sheet)

For which sample size, $n = 4$ or $n = 40$, would you trust a normal model for the sample mean? Name one feature behind your choice.

- DOK 1** (a) Identify the population shape and state the population mean and standard deviation with units.
- DOK 2** (b) For $n = 4$ and $n = 40$, calculate the mean and standard deviation of the sampling distribution of $\bar{x}$.
- DOK 3** (c) Adjudicate both claims for $n = 4$ and $n = 40$. Cite the histogram shape, independence, and the CLT's sample-size requirement; use the $n = 40$ standard deviation; and explain one consequence of ignoring skew when using a normal model at $n = 4$.

Video + follow-along: [u5_lesson3_live.html](https://www.khanacademy.org/a/u5_lesson3_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

